

This notice is issued by the NASA/Glenn Research Center (GRC) to post a Request for Information (RFI) on understanding the development needs for Special Test Equipment (STE) vacuum chambers, thermal shroud, and Thermal Conditioning Units (TCU). This RFI solicits responses from interested parties. This document is for *information and planning purposes only* while allowing industry the opportunity to comment on the reasonableness and feasibility of such equipment and whether or not there may be an interest to collaborate on such equipment. NASA would also like ROM costs and lead times for the equipment listed below. An entity does not need to be able to supply all components to respond to this RFI.

STE purpose and details:

- Vacuum Chambers 1&2 (vessel only; no pumps and associated components):
 - Approximate Test Section (inside of thermal shroud): 10' tall x 11' wide x 20' long
 - It is expected that thermal shroud will surround test section within the overall chamber shell
 - Provide approximate overall chamber size (outside dimensions) required for test section listed above
 - Rectangular or mailbox shape with a desired max height of approximately 19'.
 - Desired to accommodate up to three 60" flange OD mounted to side of chamber
 - Fully welded, non-sectional. All welds to be internal full fusion vacuum welds, unless otherwise noted
 - Max section weight (prior to final assembly): 10 tons
 - Please provide feedback if this is not possible/preferable
 - Test article weight ~4500 kg and thermal shroud (weight unknown). Chamber needs to be able to handle weight of both test article and thermal shroud
 - Provide overall approximate weight of chamber
 - 5×10^{-8} Torr capable
 - Leak rate of 5×10^{-9} STD ATM CC/SEC OR BETTER
 - #4 machine polished internal surfaces desired
 - What other options? What price differences per degree of surface finish?
 - Capable of handling temperatures down to 77 Kelvin and up to 340 Kelvin (-320F to 152F)
 - Vacuum chamber desired to be set in recessed pit in floor (See Appendix A)
 - Alternate designs that meet the other criteria would be considered
 - The key metric is that a loading floor (the location of test article in facility) must accommodate a roll in test article
 - Load floor for test equipment must accommodate a roll in test article without use of a crane or fork lift. This could be the recessed pit, a floor lift, or alternative design.
 - Door must be able to move freely from recessed pit or other design and opened with easy access to the floor level of test hardware. This should be achieved without the need for a crane or fork lift.
 - Provide other industry examples of unique door solutions especially when operating out of a recessed area
 - Removable end caps must have full coverage of interior of facility
 - Testing will involve lunar simulant soils (JSC-1/1A)
 - Materials other than those specified for use in ASME BPVC Section VIII and its reference documents may be used; however, their adequacy for the intended application

shall be demonstrated to the satisfaction of the NASA Glenn Pressure Systems Manager (PSM).

- Material Test Reports shall be submitted for the vacuum vessel.
- If a vacuum vessel requires relief protection with PRD setting above 2 psig due to the nature of the potential pressurization source, the vessel pressure shell shall be structurally qualified in accordance with the requirements of the ASME BPVC Section VIII, Div. 1.
- Vacuum vessels shall be fabricated, repaired, or altered only by either manufacturers having an ASME “Code Stamp” (ASME “U” Authorization) or NBIC “R” Stamp, or that have been audited and determined by the NASA Glenn PSM to have an equivalent quality assurance manual and process, including implementation.
- The PVS manufacturing shall be in strict accordance with the quality assurance manual of the manufacturing organization.
- All welding shall be:
 - Performed in accordance with procedures qualified in accordance with the ASME BPVC Section IX, including all essential variables for the joint in question.
 - Performed by welders qualified and current in accordance with the ASME BPVC Section IX, on such weld procedures.
- Post weld heat treatment shall be performed as required and in accordance with the ASME Code.
- Inspections shall be performed by inspectors trained and certified in use of the techniques being applied, in accordance with the requirements of the ASME BPVC Section V.
- Thermal Shroud (for Vacuum Chambers 1&2):
 - **Capable of rejecting 100 kW heat load**
 - Is a thermal shroud meeting above requirements capable to fit within Vacuum Chambers 1&2?
 - Capable of handling fully flooded Liquid Nitrogen
 - **Please provide approximate LN2 flow rate and pipe size required for 100 kW heat load**
 - Interior high emissivity paint suitable for high vacuum
 - Please describe options
 - Multi-Layer Insulation (MLI) on outside to serve as thermal barrier between thermal shroud and tank well
 - View factor should be maximized but View Factor does not need to be 1
 - What is the approximate weight of the thermal shroud of this size?
 - What type of LN2 valving is required for fully flood control?
 - What is suitable inlet and outlet flanging?
 - What is the recommended control method?
 - Capable of handling weight of test article (4500 kg) or means to pass load of test article through thermal shroud to vacuum chamber shell itself. Test article should be supported across entire length (a platform for test article to rest on).
 - Platform should allow the test article to roll on casters across its surface from outside the vacuum chamber with locations to fasten test article
 - Fully instrumented with thermocouples. Provide recommendation on distribution/spacing of thermocouples
 - This thermal shroud is desired to be removable to accommodate facility cleaning
 - Piping/tubing shall be designed, fabricated, tested and examined in accordance with ASME B31.3 (or ASME Section VIII Division 1 or 2) for a design pressure of 150 psig from 77K to 340K (–320F to 150 F).
- Vacuum Chambers 3, 4, and 5 (vessel only; no pumps and associated components):
 - Expected interior chamber size: 9’ tall x 9’ wide x 18’ long

- It is expected that thermal shroud will surround test section within the overall chamber shell
- Rectangular shape
- Modular sections desired (to allow for future length expansion)
- All welds to be internal full fusion vacuum welds, unless otherwise noted
- Max section weight (prior to final assembly): 10 tons
- Provide overall approximate weight of chamber
- 5×10^{-8} Torr capable
- Leak rate of 5×10^{-9} STD ATM CC/SEC OR BETTER
- Machine polish #4
- Capable of handling temperatures down to 77 Kelvin and up to 425 Kelvin (-320F to 306F)
- Desired to accommodate up to two 42" OD flange mounted to side of chamber
- Vacuum chamber set flat on concrete floor
- Swinging door preferred. Ideally allow opening by personnel without additional equipment.
- Testing will involve lunar simulant soils (JSC-1/1A)
- Materials other than those specified for use in ASME BPVC Section VIII and its reference documents may be used; however, their adequacy for the intended application shall be demonstrated to the satisfaction of the NASA Glenn Pressure Systems Manager (PSM).
- Material Test Reports shall be submitted for the vacuum vessel.
- If a vacuum vessel requires relief protection with PRD setting above 2 psig due to the nature of the potential pressurization source, the vessel pressure shell shall be structurally qualified in accordance with the requirements of the ASME BPVC Section VIII, Div. 1.
- Vacuum vessels shall be fabricated, repaired, or altered only by either manufacturers having an ASME "Code Stamp" (ASME "U" Authorization) or NBIC "R" Stamp, or that have been audited and determined by the NASA Glenn PSM to have an equivalent quality assurance manual and process, including implementation.
- The PVS manufacturing shall be in strict accordance with the quality assurance manual of the manufacturing organization.
- All welding shall be:
 - Performed in accordance with procedures qualified in accordance with the ASME BPVC Section IX, including all essential variables for the joint in question.
 - Performed by welders qualified and current in accordance with the ASME BPVC Section IX, on such weld procedures.
- Post weld heat treatment shall be performed as required and in accordance with the ASME Code.
- Inspections shall be performed by inspectors trained and certified in use of the techniques being applied, in accordance with the requirements of the ASME BPVC Section V.
- Thermal Shroud (for Vacuum Chambers 3, 4, and 5):
 - Capable of rejecting 1 kW heat load at minimum. What is the price difference for up to 10 kW?
 - Is a thermal shroud meeting above requirements capable to fit within Vacuum Chambers 3, 4, and 5?
 - Capable of handling fully flooded Liquid Nitrogen
 - Capable of handling heat up to 475 K (392F)
 - Interior high emissivity paint suitable for high vacuum
 - Please describe options

- Multi-Layer Insulation (MLI) on outside to serve as thermal barrier between thermal shroud and tank well
- Please provide approximate LN2 flow rate and pipe size required for 10 kW heat load
- What type of LN2 valving is required for fully flood control
- What is suitable inlet and outlet flanging
- Capable of handling weight of test article (1000 kg) or means to pass load of test article through thermal shroud to vacuum chamber shell itself. Test article should be supported across entire length (a platform for test article to rest on).
- Platform should allow the test article to roll on casters across its surface from outside the vacuum chamber with locations to fasten test article
- Fully instrumented with thermocouples
- Piping/tubing shall be designed, fabricated, tested and examined in accordance with ASME B31.3 (or ASME Section VIII Division 1 or 2) for a design pressure of 150 psig from -320 to 400 F
- Feasibility of modular thermal shrouds or thermal shrouds that can be split into zones?
- Thermal Conditioning Units (for Vacuum Chambers 3, 4, and 5)
 - Thermal Vacuum Control Systems for use with a thermal shroud in a vacuum chamber
 - These units will be controllable from 90K to 425K (-298F to 305F)
 - They will provide precise uniform control throughout the entire range
 - Will have refrigeration capacity of at least 10 kW
 - Will have heating ability of at least 10 kW
 - Will be supplied with N2 as the working fluid
 - Dynavac TCU400 up to TCU1500 or similar
 - Are other units available with similar capabilities?
 - Please provide price, capability, and lead time differences on range of sizes of TCU's

Please provide reasonableness, feasibility, and additional recommendations of such equipment and whether or not there may be an interest to supply such equipment. NASA would also like ROM costs and approximate lead times for the equipment.

NASA will not consider material that is marked classified. Any proprietary information must be clearly marked.

NASA reserves the right to use responses to share with other Federal agencies or develop future solicitations and other types of public correspondence. NASA may request to cite or release individual responses, please identify any responses that should be held confidential. Submissions must be formatted as a PDF document and limited to no more than 50 pages. Respondents are permitted to provide external links to more detailed information where appropriate.

This RFI is not to be construed as a commitment by the Government, nor will the Government pay for the information submitted in response. The Government may not respond to questions/concerns submitted. The Government will use the information to finalize the RFP as necessary, if any.

NASA may contact respondents to this RFI to seek clarifications or additional information. Responses to this RFI do not bind NASA to any further actions related to this topic. Any future steps taken to potentially award Broad Agency Announcements, Request for Proposals, Space Act Agreements, or Announcements of Opportunities will be contingent upon availability of funds. This announcement contains all information required to submit a response.

All comments or questions shall be submitted electronically via email to the Contracting Officer, Paige Foreman at paige.e.foreman@nasa.gov, no later than April 27, 2026, 3:00PM EST. When responding reference 80GRC026RFI01.

NASA Clause 1852.215-84, Ombudsman, is applicable. The Center Ombudsman for this acquisition can be found at <https://www.hq.nasa.gov/office/procurement/regs/Procurement-Ombuds-Comp-Advocate-Listing.pdf>.

If a solicitation is released, then it and any additional documents will be available on www.sam.gov. It is the responsibility of the respondents to monitor this website for the release of the solicitation and amendments (if any). Potential responders will be responsible for downloading their own copy of the solicitation and amendments, if any.

Appendix A – Conceptual Diagram of Vacuum Chambers 1&2 in Recessed Pit.

